

NGUYEN DINH NGHIA

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PROFESSIONAL PROFILE

Solution Architect with 10+ years of experience designing and implementing enterprise-scale systems (ERP, CRM, Data Platforms) serving millions of users. Expert in system architecture, data engineering, and modernizing legacy systems into Event-Driven and Microservices architectures. Proven track record in building high-performance solutions and establishing technical standards across organizations.

TECHNICAL EXPERTISE

Architecture & Design: Microservices, Event-Driven Architecture (EDA), System Design (C4 Model), Data Lakehouse Architecture, API Gateway, Multi-tenancy, Master Data Management (MDM)

Data Engineering: Hadoop, Spark, Hudi, Presto, ClickHouse, ETL Pipelines, CDC, Real-time Analytics

Backend & Databases: C#/.NET Core, Python, SQL, JavaScript | SQL Server (Performance Tuning Expert), PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

DevOps & Cloud: Docker, Kubernetes, Kafka, RabbitMQ, Airflow, Prometheus, Grafana, OpenTelemetry, Distributed Tracing

Methodology: Agile/Scrum, Domain-Driven Design (DDD), CQRS, Event Sourcing

PROFESSIONAL EXPERIENCE

Principal Solution Architect / Data Engineering Lead | MISA JSC

Jun 2023 - Present

Outline:

Lead architect for data engineering and analytics, managing team of 8+ engineers responsible for company-wide data infrastructure and strategic insights.

Key Responsibilities:

- Built and led data engineering team establishing technical standards and best practices
- Designed and architected Data Lakehouse platform (Hadoop, Hudi, Presto) managing 20TB+ of data for 2M+ enterprise clients
- Optimized ClickHouse data warehouse reducing complex report generation time by 80%
- Architected robust ETL pipelines for automated data ingestion from 50+ disparate sources
- Deployed AI/ML predictive models for churn analysis and customer segmentation
- Delivered real-time and periodic reports to executive board on subscriber trends and business metrics

Technology: Python, C#/.NET Core, Hadoop, Spark, Hudi, Presto, ClickHouse, Kafka, PostgreSQL, Kubernetes, Airflow, OpenTelemetry

Key Achievements: - Reduced data processing time by 80% enabling faster business insights for C-level executives - Integrated 50+ disparate data sources into unified platform serving 2M+ enterprise clients - Improved system observability reducing incident response time by 60% through distributed tracing

Principal Solution Architect / Database Expert | MISA JSC

Jan 2020 - May 2023

Outline:

Lead architect responsible for company-wide system architecture, technical standards, and complex

solution design. Mentored DBAs across all product lines.

Key Responsibilities:

- Re-architected E-invoice system from monolith to microservices, scaling from 100K to 1M invoices/day
- Led architecture and development of comprehensive ERP system serving 100K+ enterprise clients (Accounting, HRM, CRM modules)
- Implemented Event-Driven Architecture with distributed tracing (OpenTelemetry) to improve system observability
- Diagnosed and resolved critical performance bottlenecks achieving 5x to 100x improvements
- Authored company-wide technical guidelines for database architecture, API design, and system implementation
- Mentored development teams and DBAs on best practices for high-performance system design

Technology: C#/.NET Core, Python, SQL Server, PostgreSQL, MongoDB, Kafka, Kubernetes, Redis, Elasticsearch, OpenTelemetry

Key Achievements: - Scaled e-invoicing system 10x (100K to 1M invoices/day) supporting national digital transformation - Optimized critical HRM reports from 10s to <1s response time (10x improvement) - Reduced data processing time by 80% through ClickHouse optimization - Established database and architecture standards adopted across 40+ applications - Reduced production incidents by 70% through proactive performance optimization

Tech Lead / Solution Architect | MISA JSC

Jan 2018 - Dec 2019

Outline:

Led technical design and architecture for flagship ERP/CRM platform serving 100K+ enterprise customers.

Key Responsibilities:

- Architected scalable CRM platform supporting 2,000+ sales staff and managing 2M+ customer records
- Designed system integration architecture connecting 40+ internal applications
- Led technical decisions on technology stack, patterns, and best practices
- Optimized system performance for business-critical modules (HRM, invoicing, CRM)
- Mentored development team on architecture patterns and code quality

Technology: C#/.NET Core, MySQL, Kafka, RabbitMQ, Elasticsearch, Kubernetes, Angular

Key Achievements: - Designed architecture that became company's most successful product - Achieved zero data loss during migration of 2M+ customer records from legacy system - Delivered scalable solution handling millions of transactions daily

Software Engineer / Technical Lead | FPT Software

Jun 2012 - Jan 2017

Outline:

Developed mission-critical enterprise systems for Japanese clients (65,000+ organizations).

Key Responsibilities:

- Designed and developed Email Management System with Exchange Server integration
- Built Outlook add-in with complex contact data management
- Performed SQL Server performance tuning and query optimization
- Designed relational database schemas for enterprise applications
- Collaborated with international clients on technical specifications and system design

Technology: C#/.NET Framework, SQL Server, Exchange Server, Windows Forms, WPF

Key Achievements: - Delivered enterprise systems with 99.9% uptime for 65,000+ organizations - Recognized for technical excellence in Exchange Server integration and database optimization

EDUCATION

Hanoi University of Science and Technology | Oct 2007 - May 2012
Major: Applied Mathematics and Informatics

HONORS & RECOGNITION

Editor's Choice - Viblo (January 2023)
Awarded for outstanding technical contributions to the developer community